

```

Model{

#####
# FIRST PART      #
#####

#####
# 1.CALIBRATION #
#####

d_moy~dgamma(1,0.001)
beta_d~dgamma(0.001,0.001)

inv_kappa ~ dgamma(0.001,0.001)
kappa <-1/ inv_kappa

#eta ~ dgamma(0.001,0.001)

L_sigma_p~dunif(0.0001,10)
L_tau_p<-pow(L_sigma_p,-2)
L_var_p<-1/L_tau_p

L_mu_p ~dnorm(0,0.001)
logit(mu_p)<-L_mu_p

alpha_d<-d_moy * beta_d

for (g in 1:calib){

#=====
# 1.1 Density part
#=====
d[g]~dgamma(alpha_d,beta_d)
lambda_IA[g]<-kappa*d[g]      #kappa*pow(d[g],eta)
EF_IA[g]~dpois(lambda_IA[g])
lambda_N[g]<-(d[g]*S[g])-EF_IA[g]

#Abundance follows a Poisson distribution
N_tot[g]~dpois(lambda_N[g])I(,2000)

L_p[g]~dnorm(L_mu_p,L_tau_p)
logit(p[g]) <- L_p[g]

# depletion pass part
C_1[g]~dbin(p[g],N_tot[g])
N_1[g]<-N_tot[g]-C_1[g]
C_2[g]~dbin(p[g],N_1[g])

#=====
# 1.2 Posterior check
#=====
rep_lambda_IA[g]<-kappa*d[g]      #kappa*pow(d[g],eta)
rep_EF_IA[g]~dpois(rep_lambda_IA[g])
res_IA_EF[g]<-EF_IA[g] - rep_EF_IA[g]

# calculation of the residuals for the predicted C1,C2,C3 conditionally to densities

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```

rep_C_1[g] ~ dbin(p[g],N_tot[g])
rep_C_2[g] ~ dbin(p[g],N_1[g])

res_C_1[g] <- C_1[g]-rep_C_1[g]
res_C_2[g] <- C_2[g]-rep_C_2[g]
res_C1_N_tot[g] <- (C_1[g]-rep_C_1[g])/N_tot[g]
res_C_2_N_1[g] <- (C_2[g]-rep_C_2[g])/N_1[g]

}

#=====
# 1.3 cut of all the parameters of the calibration
#=====
L_mu_p_cut<-cut(L_mu_p)
L_tau_p_cut<-cut(L_tau_p)
kappa_cut<-cut(kappa)
#eta_cut<-cut(eta)
# on suppose que le parametre d'echelle est le meme au fil des années
beta_d_cut<-cut(beta_d)

#####
# 2. REDD/SPAWNERS #
#####
#=====
# 2.1 Parameters of the Redd/spawner relationship model
#=====
mu_zone[1]~dgamma(1,0.001)
mu_zone[2]<-1      #-dgamma(1,0.001)

beta_zone~dgamma(0.01,0.01)

alpha_zone[1]<- mu_zone[1]*beta_zone

alpha_zone[2]<- mu_zone[2]*beta_zone

#=====
# 2.2 Methodology and spatial effect
#=====
#-----
# 2.2.1 Methodology effect
#-----
hel_effect[1]<-1
hel_effect[2]~dgamma(1,1)
#-----
# 2.2.2 Spatial effect
#-----
for (t in 1:T){
  zone_effect[t,1]~dgamma(alpha_zone[1],beta_zone)l(0.001,)
  zone_effect[t,2]~dgamma(alpha_zone[1],beta_zone)l(0.001,)
  zone_effect[t,3]~dgamma(alpha_zone[1],beta_zone)l(0.001,)
}
for (t in 12:T){
  zone_effect[t,4]~dgamma(alpha_zone[2],beta_zone)l(0.001,)
}

```

```

#-----
# 2.2.3 Verification of both effects
#-----
diff_hel_effect<-1-hel_effect[2]
P_diff_hel<-step(diff_hel_effect)

diff_zone1_2<-mu_zone[1]-mu_zone[2]
p_diff_zone1_2<-step(diff_zone1_2 )

#=====
# 2.3 Area prospected
#=====
#loops for proportion of area prospected
for (t in 1:T){
  for (k in 1:3){
    logit(p_area[t,k])<- L_p_area[t,k]
  }
}

for (t in 12:T){
  logit(p_area[t,4])<- L_p_area[t,4]
}

#=====
# 2.4 Hyperparameters
#=====
sigma_Vichy <- sqrt( 1 / tau_vichy)   #~dunif(0.001,5)
tau_vichy~dgamma(0.001,0.001)        #<-pow(sigma_Vichy,-2)
L_mu_vichy~dnorm(0,0.01)

      sigma_p_alagnon<-sqrt(1/tau_p_alagnon)
      tau_p_alagnon~dgamma(0.01,0.01)

sigma_p_langeac<-sqrt(1/tau_p_langeac)
tau_p_langeac~dgamma(0.01,0.01)

sigma_p_poutes<-sqrt(1/tau_p_poutes)
tau_p_poutes~dgamma(0.01,0.01)

I_surv[7] <- 1
I_surv_prim[7] <- 1

level_s~dnorm(0,1)

for (t in 8:T){
  I_surv_prim[t] ~ dbern(0.5)
  I_surv[t] <- I_surv[t-1] * I_surv_prim[t]
}

for (t in 7:11){
  min_N_1[t]<-tot_C[t] + S_stocking[t]+2
  pool_juv[t]<-s_juv2ad* Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 * smolts_tot[t] )
  L_mu_Vichy_nm[t]<-log( s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 *
smolts_tot[t] )) + level_s * I_surv[t]
  mean_y_surv[t] <- s_juv2ad * exp(level_s * I_surv[t])
}

```

```

N[t,1]~dlnorm(L_mu_Vichy_nm[t],tau_vichy)l(min_N_1[t],15000)
res_Vichy[t] <- log(N[t,1]) - L_mu_Vichy_nm[t]
}

```

```
#####
```

```
# 3. JUVENILE PRODUCTION #
```

```
#####
```

```
#=====
```

```
# 3.1 Beverthon & Holt parameters
```

```
#=====
```

```
# BH slope parameter
```

```
#not sure about the beta parameters ...
```

```
zt~dbeta(1,9)#(1,2)
```

```
a<-zt *8000 #l~dunif(1,8000) #~dgamma(0.01,0.01)l(,8000)#(0.1,) #~dlnorm(0,0.01) #
```

```
alpha_dd<- 1/a
```

```
a_juv ~dbeta(2,2)
```

```
alpha_dd_juv<- 1/a_juv
```

```
Rmax~dunif(0,2)
```

```
beta_dd<- 1 / Rmax
```

```
s_egg~dbeta(2,2)
```

```
s_juv2ad~dbeta(2,2)
```

```
#=====
```

```
# 3.2 Hyperparameters
```

```
#=====
```

```
alpha_tau <- mu_tau +1
```

```
mu_tau ~ dgamma(0.1,0.1)l(0.000001,)
```

```
beta_tau ~ dgamma(0.1,0.1)l(0.001,)
```

```
tau_wild_moy~dgamma(alpha_tau,beta_tau)
```

```
tau_wild_site~dgamma(alpha_tau,beta_tau)
```

```
tau_juv_moy[1]~dgamma(0.01,0.01)
```

```
tau_juv_site[1]~dgamma(0.01,0.01)
```

```
tau_juv_moy[2]~dgamma(alpha_tau,beta_tau)
```

```
tau_juv_site[2]~dgamma(alpha_tau,beta_tau)
```

```
tau_egg_moy[1]~dgamma(0.01,0.01)l(0.01,)
```

```
tau_egg_site[1]~dgamma(0.01,0.01)l(0.01,)
```

```
tau_egg_moy[2]~dgamma(alpha_tau,beta_tau)#l(,50)
```

```
tau_egg_site[2]~dgamma(alpha_tau,beta_tau)
```

```
sigma_wild_moy <- sqrt( 1 / tau_wild_moy)
```

```
sigma_wild_site <- sqrt( 1 / tau_wild_site)
```

```
sigma_juv_moy <- sqrt( 1 / tau_juv_moy[2])
```

```
sigma_juv_site <- sqrt( 1 / tau_juv_site[2])
```

```
sigma_egg_moy <- sqrt( 1 / tau_egg_moy[2])
```

```
sigma_egg_site <- sqrt( 1 / tau_egg_site[2])
```

```
nu_wild_avg~dnorm(0,0.01)
```

```
nu_wild[1] <- -nu_wild_avg
```

```
nu_wild[2] <- nu_wild_avg
```

```

nu_wild[3] <- nu_wild_avg
nu_wild[4] <- nu_wild_avg

rho_poutes~dbeta(2,2)
#=====
# 3.3 Number of juveniles 0+ returning in the Allier river for a given year
#=====
# 0+ Juvenile returning in the Allier for a given year
# and originating from the 3 areas of interest
for (t in 7:T){
  Juv_tot[t,1] <- (1/3) * Juv[t-3,1] + (1/3) * Juv[t-4,1] + (1/3) * Juv[t-5,1]
  Juv_tot[t,2] <- (1/3) * Juv[t-3,2] + (1/3) * Juv[t-4,2] + (1/3) * Juv[t-5,2]
  Juv_tot[t,3] <- (1/3) * Juv[t-3,3] + (1/3) * Juv[t-4,3] + (1/3) * Juv[t-5,3]
}
  for (t in 1:15){
    Juv_tot[t,4]<-0
  }
for (t in 16:16){
  Juv_tot[t,4] <- (1/3) * Juv[t-3,4]
}
for (t in 17:17){
  Juv_tot[t,4] <- (1/3) * Juv[t-3,4] + (1/3) * Juv[t-4,4]
}
for (t in 18:T){
  Juv_tot[t,4] <- (1/3) * Juv[t-3,4] + (1/3) * Juv[t-4,4] + (1/3) * Juv[t-5,4]
}
#   for (t in 30:30){
#     Juv_tot[t,2]<-1/3*Juv[t-3,2]
#   }
#   for (t in 31:31){
#     Juv_tot[t,2]<-1/3*Juv[t-3,2] + 1/3*Juv[t-4,2]
#   }
#   for (t in 32:T){
#     Juv_tot[t,2]<-1/3*Juv[t-3,2] + 1/3*Juv[t-4,2] + 1/3*Juv[t-5,2]
#   }
for (t in 7:15){
  Juv_tot_system[t] <- Juv_tot[t,1]+ Juv_tot[t,2] +Juv_tot[t,3]
}
for (t in 16:T){
  Juv_tot_system[t] <- Juv_tot[t,1]+Juv_tot[t,2] +Juv_tot[t,3] +rho_poutes*Juv_tot[t,4]
}
# for (t in 30:T){
#   Juv_tot_system[t] <- Juv_tot[t,1] + Juv_tot[t,2] +Juv_tot[t,3] + rho_poutes*Juv_tot[t,4]
# }

#=====
# 3.4 Probability of passing at Vichy, Alagnon, Langeac and Poutes
#=====
# incorporating the effect that probability of passing at Langeac and Poutes is conditioned by the
amount of juvenile produced

#Probability to reach Vichy if not catch downstream
p_reach_V~dbeta(2,1)
for (t in 1:T){
  C_dwn_reach[t] <- p_reach_V * C_dwn[t]
}

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    tot_C[t] <- round( C_dwn_reach[t] + C_up[t])
  }
  for (t in 1:6){
    min_N_1[t] <- tot_C[t] + S_stocking[t]+2
    N[t,1]~dlnorm(6.9,0.0453)(min_N_1[t],15000)
  }

  # For Alagnon, Langeac et Poutes : filter:
  # if negative : fish returning in smaller proportion than what was expected regarding juvenile
  production
  # if positive : fish returning in higher proportion than what was expected regarding juvenile production
  adjust_p_L ~ dnorm(0,0.01)
  adjust_p_P ~ dnorm(0,0.01)
  adjust_p_A ~ dnorm(0,0.01)
  rho_station~dbeta(2,2)

  for (t in 1:4){

    ratio_juv_prod_A[t]~dbeta(1,3)
    ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3])) + (1-
rho_station) * ratio_juv_prod_A[t]
    L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
    L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
    res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

    ratio_juv_prod_L[t]~dbeta(3,1)
    ratio_juv_L[t]<- rho_station * (S_juv_JP[t,3]/(S_juv_JP[t,1]+S_juv_JP[t,3])) + (1 - rho_station) *
ratio_juv_prod_L[t]
    L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
    L_mu_p_langeac[t]<-L_ratio_juv_L[t] + adjust_p_L
    L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
    res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]
  }
  for (t in 5:5){

    ratio_juv_prod_A[t]<- Juv[t-3,2] / ( Juv[t-3,1] + Juv[t-3,2]+ Juv[t-3,3])
    ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3])) + (1-
rho_station) * ratio_juv_prod_A[t]
    L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
    L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
    res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

    ratio_juv_prod_L[t] <- Juv[t-3,3] / ( Juv[t-3,1] + Juv[t-3,3] )
    ratio_juv_L[t]<- rho_station * (S_juv_JP[t,3]/(S_juv_JP[t,1]+S_juv_JP[t,3])) + (1 - rho_station) *
ratio_juv_prod_L[t]
    L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
    L_mu_p_langeac[t]<-L_ratio_juv_L[t]+ adjust_p_L
    L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
    res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]
  }
  for (t in 6:6){

    ratio_juv_prod_A[t]<- (Juv[t-3,2]+Juv[t-4,2]) / ( Juv[t-3,1] +Juv[t-4,1]+ Juv[t-3,2] +Juv[t-4,2]+ Juv[t-
3,3]+Juv[t-4,3])

```

```

ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3])) + (1-
rho_station) * ratio_juv_prod_A[t]
L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

ratio_juv_prod_L[t] <- (Juv[t-3,3] + Juv[t-4,3]) / ( Juv[t-3,1] + Juv[t-4,1] + Juv[t-3,3] +Juv[t-4,3] )
ratio_juv_L[t]<- rho_station * (S_juv_JP[t,3]/(S_juv_JP[t,1]+S_juv_JP[t,3])) + (1 - rho_station) *
ratio_juv_prod_L[t]
L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
L_mu_p_langeac[t]<-L_ratio_juv_L[t]+ adjust_p_L
L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]
}
for (t in 7:11){

ratio_juv_prod_A[t]<- Juv_tot[t,2] / ( Juv_tot[t,1] + Juv_tot[t,2]+ Juv_tot[t,3] )
ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3]))+ (1-
rho_station) * ratio_juv_prod_A[t]
L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

ratio_juv_prod_L[t] <- Juv_tot[t,3] / ( Juv_tot[t,1] + Juv_tot[t,3] )
ratio_juv_L[t]<- rho_station * (S_juv_JP[t,3]/(S_juv_JP[t,1]+S_juv_JP[t,3])) + (1 - rho_station) *
ratio_juv_prod_L[t]
L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
L_mu_p_langeac[t]<-L_ratio_juv_L[t]+ adjust_p_L
L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]
}
for (t in 12:15){

ratio_juv_prod_A[t]<- Juv_tot[t,2] / ( Juv_tot[t,1] + Juv_tot[t,2]+ Juv_tot[t,3]+ Juv_tot[t,4]
*rho_poutes)
ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3]+
S_juv_JP[t,4])) + (1-rho_station) * ratio_juv_prod_A[t]
L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

ratio_juv_prod_L[t] <- Juv_tot[t,3] / ( Juv_tot[t,1] + Juv_tot[t,3]+ Juv_tot[t,4] *rho_poutes )
ratio_juv_L[t]<- rho_station *
((S_juv_JP[t,3]+S_juv_JP[t,4])/((S_juv_JP[t,1]+S_juv_JP[t,3]+S_juv_JP[t,4])) + (1 - rho_station) *
ratio_juv_prod_L[t]
L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
L_mu_p_langeac[t]<-L_ratio_juv_L[t]+ adjust_p_L
L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]

ratio_juv_prod_P[t] <- 0
ratio_juv_P[t]<- rho_station * (S_juv_JP[t,4] / (S_juv_JP[t,3] + S_juv_JP[t,4]) ) + (1 - rho_station) *
ratio_juv_prod_P[t]
L_ratio_juv_P[t] <- logit(ratio_juv_P[t])
L_mu_p_poutes[t]<-L_ratio_juv_P[t] + adjust_p_P
L_p_poutes[t]~dnorm(L_mu_p_poutes[t],tau_p_poutes)

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    res_p_poutes[t] <- L_p_poutes[t] - L_mu_p_poutes[t]
  }
  for (t in 16:T){

    ratio_juv_prod_A[t]<- Juv_tot[t,2] / ( Juv_tot[t,1] + Juv_tot[t,2]+ Juv_tot[t,3]+ Juv_tot[t,4]
*rho_poutes)
    ratio_juv_A[t]<-rho_station * (S_juv_JP[t,2]/(S_juv_JP[t,1]+S_juv_JP[t,2]+S_juv_JP[t,3]+
S_juv_JP[t,4])) + (1-rho_station) * ratio_juv_prod_A[t]
    L_mu_p_alagnon[t]<-logit(ratio_juv_A[t]) + adjust_p_A
    L_p_alagnon[t]~dnorm(L_mu_p_alagnon[t],tau_p_alagnon)
    res_p_alagnon[t]<-L_p_alagnon[t]-L_mu_p_alagnon[t]

    ratio_juv_prod_L[t] <- (Juv_tot[t,3] + Juv_tot[t,4]*rho_poutes) / ( Juv_tot[t,1] + Juv_tot[t,3] +
Juv_tot[t,4]*rho_poutes )
    ratio_juv_L[t]<-
rho_station*((S_juv_JP[t,3]+S_juv_JP[t,4])/(S_juv_JP[t,1]+S_juv_JP[t,3]+S_juv_JP[t,4])) + (1 -
rho_station) * ratio_juv_prod_L[t]
    L_ratio_juv_L[t] <- logit(ratio_juv_L[t])
    L_mu_p_langeac[t]<-L_ratio_juv_L[t]+ adjust_p_L
    L_p_langeac[t]~dnorm(L_mu_p_langeac[t],tau_p_langeac)
    res_p_langeac[t] <- L_p_langeac[t] - L_mu_p_langeac[t]

    ratio_juv_prod_P[t] <- Juv_tot[t,4] *rho_poutes/ ( Juv_tot[t,3] + Juv_tot[t,4] *rho_poutes)
    ratio_juv_P[t]<- rho_station * (S_juv_JP[t,4] / (S_juv_JP[t,3] + S_juv_JP[t,4])) + (1 - rho_station) *
ratio_juv_prod_P[t]
    L_ratio_juv_P[t] <- logit(ratio_juv_P[t])
    L_mu_p_poutes[t]<-L_ratio_juv_P[t]+ adjust_p_P
    L_p_poutes[t]~dnorm(L_mu_p_poutes[t],tau_p_poutes)
    res_p_poutes[t] <- L_p_poutes[t] - L_mu_p_poutes[t]
  }
}

```

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#####
# SECONDE PART #
#####

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```

#####
# 1. LOOP FOR YEARS (only downstream Poutès) #
#####

```

```

for (t in 1:11){
  #=====
  # 1.1 Redd/Spawners part
  #=====
  logit(p_alagnon[t])<-L_p_alagnon[t]
  logit(p_langeac[t])<- L_p_langeac[t]
  max_N_langeac[t]<- N_corr[t] - 1

  #without fish caught for breeding or rod catches
  N_corrected[t] <- N[t,1] - tot_C[t] - S_stocking[t]
  N_corr[t]<-N_corrected[t]-N[t,2]
  N[t,2]~dbin(p_alagnon[t],N_corrected[t])
  N[t,3]~dbin(p_langeac[t],N_corr[t])
  #-----
  #1.1.1 Number of potential spawners
  #-----
  S_ts[t,1]<- max( N[t,1] - N[t,2] - N[t,3] - tot_C[t] - S_stocking[t],1)
}

```



```

S_ts[t,2]<- max(N[t,2],1)
S_ts[t,3]<-max(N[t,3],1)

#ratio_S[t,1] <- S_ts[t,1] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3])
#ratio_S[t,2] <- S_ts[t,2] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3])
#ratio_S[t,3] <- S_ts[t,3] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3])

#####
# 1.2 Loop for zones (1= Vichy-Langeac, 2= Alagnon, 3= Langeac-Poutès, 4= upstream Poutès)
#####

for (i in 1:3){
#-----
# 1.2.1 Redd/Spawners part
#-----
#.....
# 1.2.1.1 estimation of the spawners
#.....
R[t,i] ~dpois(lambda[t,i])
lambda[t,i] <- S_ts[t,i] *zone_effect[t,i] * hel_effect[1] *p_area[t,i]

#residus calculés pour êtres centrés sur 0 avec varaince homogene
res_R[t,i]<-(R[t,i]-lambda[t,i])/sqrt(lambda[t,i])
#.....
# 1.2.1.2 Cut of all parameters
#.....
lambda_cut[t,i]<-cut(lambda[t,i])
R_rep[t,i]~dpois(lambda_cut[t,i])
#-----
# 1.2.2 Juvenile production
#-----
# l_juv_moy = indicator for stocking of 0+ or not
# l_egg_moy = indicator for stocking of eggs or not
#d_tot_moy without taking into account area for the stocked juveniles (data only from year 31)
d_tot_moy[t+1,i] <- d_wild_moy[t+1,i] + l_juv_moy[t+1,i] * d_juv_moy[t+1,i]
Juv[t+1,i] <- d_tot_moy[t+1,i]*S_juv_JP[t+1,i]
#.....
# 1.2.2.1 Wild component
#.....
log(d_wild_moy[t+1,i]) <- L_d_wild_moy[t+1,i]
L_d_wild_moy[t+1,i] ~ dnorm(L_mu_d_wild[t+1,i],tau_wild_moy)I(-6.91,1.09)
L_mu_d_wild[t+1,i] <- log((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd * (S_ts[t,i]/S_juv_JP[t,i]))) +
nu_wild[i]
res_wild_moy[t+1,i] <- L_d_wild_moy[t+1,i] - L_mu_d_wild[t+1,i]
#.....
# 1.2.2.2 stocked juvenile component
#.....
log(d_juv_moy[t+1,i]) <- L_d_juv_moy[t+1,i]
L_d_juv_moy[t+1,i] ~ dnorm(L_mu_d_juv[t+1,i],tau_juv_moy[ l_juv_moy[t+1,i]+1])I(-6.91,1.09)
#<- L_mu_d_juv[t+1,i] #
Rmax_juv_temp[t+1,i] <- ( Rmax - ((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd *
(S_ts[t,i]/S_juv_JP[t,i])) ) * exp(nu_wild[i])
Rmax_juv[t+1,i] <- max(Rmax_juv_temp[t+1,i] ,0.000001)
beta_dd_juv[t+1,i] <- 1 / Rmax_juv[t+1,i]

```

```

L_mu_d_juv[t+1,i] <- I_juv_moy[t+1,i] * log( (stock_juv[t+1,i]/S_juv_JP[t+1,i]) /
(alpha_dd_juv/exp(nu_wild[i]) + beta_dd_juv[t+1,i] * (stock_juv[t+1,i]/S_juv_JP[t+1,i])))
res_juv_moy[t+1,i] <- L_d_juv_moy[t+1,i] - L_mu_d_juv[t+1,i]
# getting out of the zone loop, one loop for each zones and the local densitie
# to avoid using 3 dimensions matrix
}
#-----
# 1.2.3 Successive removal fisheries
#-----
# I_site_juv_V/L/P = indicator for presence/absence of stocking on the site
# loop for sites with successive removal EF (DE LURY)
# Pas de pêche Delury pour l'Alagnon
#.....
# 1.2.3.1 zone 1 : Vichy Langeac
#.....
for (k in 1:J[t+1,1]){
  d_V[t+1,k] <- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k]
  log(d_wild_V[t+1,k]) <- L_d_wild_V[t+1,k]
  L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,1.09)
  log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
  L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[ I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)

  lambda_N_V[t+1,k] <- d_V[t+1,k]*S_depl_V[t+1,k]
  #Abundance follows a Poisson distribution
  N_tot_V[t+1,k] ~ dpois(lambda_N_V[t+1,k])

  L_p_V[t+1,k] ~ dnorm(L_mu_p_cut,L_tau_p_cut)
  logit(p_V[t+1,k]) <- L_p_V[t+1,k]

  C_1_V[t+1,k] ~ dbin(p_V[t+1,k],N_tot_V[t+1,k])
  N_1_V[t+1,k] <- N_tot_V[t+1,k]-C_1_V[t+1,k]
  #not all sites have 2 pass, this vector show which sites does
  for (h in 1:pass_2_V[t+1,k]){
    C_2_V[t+1,k] ~ dbin(p_V[t+1,k],N_1_V[t+1,k])
    N_2_V[t+1,k] <- N_1_V[t+1,k]-C_2_V[t+1,k]
  }
}

#.....
# 1.2.3.3 zone 3 : Langeac Poutes
#.....
for (k in 1:J[t+1,3]){
  d_L[t+1,k] <- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k]
  log(d_wild_L[t+1,k]) <- L_d_wild_L[t+1,k]
  L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,1.09)
  log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
  L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

  lambda_N_L[t+1,k] <- d_L[t+1,k]*S_depl_L[t+1,k]
  #Abundance follows a Poisson distribution
  N_tot_L[t+1,k] ~ dpois(lambda_N_L[t+1,k])

  L_p_L[t+1,k] ~ dnorm(L_mu_p_cut,L_tau_p_cut)
  logit(p_L[t+1,k]) <- L_p_L[t+1,k]

```

```

C_1_L[t+1,k]~dbin(p_L[t+1,k],N_tot_L[t+1,k])
N_1_L[t+1,k]<-N_tot_L[t+1,k]-C_1_L[t+1,k]
#not all sites have 2 pass, this vector show which sites does
for (h in 1:pass_2_L[t+1,k]){
  C_2_L[t+1,k]~dbin(p_L[t+1,k],N_1_L[t+1,k])
  N_2_L[t+1,k]<-N_1_L[t+1,k]-C_2_L[t+1,k]
  #not all sites have 3 pass, this vector show which sites does
  for (m in 1:pass_3_L[t+1,k]){
    C_3_L[t+1,k]~dbin(p_L[t+1,k],N_2_L[t+1,k])
  }
}
}
}

```

```

#####
####

```

```

# 2. LOOP FOR YEARS (all zones mais pas encore de juvéniles à Poutès - seulement en année T=16)
#

```

```

#####
####

```

```

for (t in 12:22){
  #=====
  # 2.1 Redd/Spawners part
  #=====
  logit(p_alagnon[t])<-L_p_alagnon[t]
  logit(p_langeac[t])<- L_p_langeac[t]
  logit(p_poutes[t])<- L_p_poutes[t]
  pool_juv[t]<-s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 * smolts_tot[t] )
  L_mu_Vichy_nm[t]<-log(s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 *
smolts_tot[t] )) + level_s * l_surv[t]
  mean_y_surv[t] <- s_juv2ad * exp(level_s * l_surv[t])

```

```

min_N_1[t]<-max(N[t,4]+2,tot_C[t] +2)+S_stocking[t]
N[t,1]~dlnorm(L_mu_Vichy_nm[t],tau_vichy)l(min_N_1[t],15000)
#without fish caught for breeding or rod catches
N_corrected[t] <- N[t,1] - tot_C[t] - S_stocking[t]
N_corr[t]<-N_corrected[t] - N[t,2]
res_Vichy[t] <- log(N[t,1]) - L_mu_Vichy_nm[t]

```

```

max_N_langeac[t]<- N_corr[t] - 1
min_L_P[t]<-max(min_L[t], N[t,4]+1)
N[t,2]~dbin(p_alagnon[t],N_corrected[t]) #Pas de l car on pourrait avoir 0 sur alagnon et tout sur
allier une année donnée
N[t,3]~dbin(p_langeac[t],N_corr[t])l(min_L_P[t],)

```

```

max_N_poutes[t]<-N[t,3]-1
N[t,4]~dbin(p_poutes[t],N[t,3])
#-----
# 2.1.1 Number of potential spawners
#-----
#mu_S_ts[t,1]<- N[t,1] - N[t,2] - N[t,3] - S_stocking[t]
#mu_S_ts[t,2]<- N[t,2]
#mu_S_ts[t,3]<- N[t,3]-N[t,4]

```

```

#test[t]<-mu_S_ts[t,1]-S_ts[t,1]

S_ts[t,1]<- max(N[t,1] - N[t,2] - N[t,3] - S_stocking[t] - tot_C[t] ,1)
S_ts[t,2]<- max( N[t,2],1)
S_ts[t,3]<- max( N[t,3]-N[t,4],1)
S_ts[t,4]<-max( N[t,4],1)
#ratio_S[t,1] <- S_ts[t,1] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,2] <- S_ts[t,2] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,3] <- S_ts[t,3] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,4] <- S_ts[t,4] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])

#=====
# 2.2 Loop for zones (1= Vichy-Langeac, 2= Alagnon, 3= Langeac-Poutès, 4= upstream Poutès)
#=====

for (i in 1:4){
#-----
# 2.2.1 Redd/Spawners part
#-----
#.....
# 2.2.1.1 estimation of the spawners
#.....
R[t,i]~dpois(lambda[t,i])
lambda[t,i] <- S_ts[t,i] *zone_effect[t,i]* hel_effect[1] *p_area[t,i]
res_R[t,i]<-(R[t,i]-lambda[t,i])/sqrt(lambda[t,i])
#.....
# 2.2.1.2 Cut of all parameters
#.....
lambda_cut[t,i]<-cut(lambda[t,i])
R_rep[t,i]~dpois(lambda_cut[t,i])

#-----
# 2.2.2 Juvenile production
#-----
# l_juv_moy = indicator for stocking of 0+ or not
# l_egg_moy = indicator for stocking of eggs or not
#d_tot_moy without taking into account area for the stocked juveniles (data only from year 31)
d_tot_moy[t+1,i] <- d_wild_moy[t+1,i] + l_juv_moy[t+1,i] * d_juv_moy[t+1,i] + l_egg_moy[t+1,i] *
d_egg_moy_surf[t+1,i]
Juv[t+1,i] <- d_tot_moy[t+1,i]*S_juv_JP[t+1,i]
d_egg_moy_surf[t+1,i] <- d_egg_moy[t+1,i]
#.....
# 2.2.2.1 Wild component
#.....
log(d_wild_moy[t+1,i]) <- L_d_wild_moy[t+1,i]
L_d_wild_moy[t+1,i] ~ dnorm(L_mu_d_wild[t+1,i],tau_wild_moy)I(-6.91,1.09) #<-
L_mu_d_wild[t+1,i] #
L_mu_d_wild[t+1,i] <- log((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd * (S_ts[t,i]/S_juv_JP[t,i]))) +
nu_wild[i]
res_wild_moy[t+1,i] <- L_d_wild_moy[t+1,i] - L_mu_d_wild[t+1,i]
#.....
# 2.2.2.2 stocked juvenile component
#.....
log(d_juv_moy[t+1,i]) <- L_d_juv_moy[t+1,i]
L_d_juv_moy[t+1,i] ~ dnorm(L_mu_d_juv[t+1,i],tau_juv_moy[ l_juv_moy[t+1,i]+1])I(,1.09)

```

```

# We recalculate the Rmax "available" to stocked 0+ by subtracting wild 0+ density and stocked
eggs density
# to the total Rmax of the density dependence relationship
Rmax_juv_temp[t+1,i] <- ( Rmax - ((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd *
(S_ts[t,i]/S_juv_JP[t,i])) ) * exp(nu_wild[i])
Rmax_juv[t+1,i] <- max( Rmax_juv_temp[t+1,i] , 0.000001)
beta_dd_juv[t+1,i] <- 1 / Rmax_juv[t+1,i]
L_mu_d_juv[t+1,i] <- L_juv_moy[t+1,i] * log( (stock_juv[t+1,i]/S_juv_JP[t+1,i]) /
(alpha_dd_juv/exp(nu_wild[i]) + beta_dd_juv[t+1,i]*(stock_juv[t+1,i]/S_juv_JP[t+1,i])))
res_juv_moy[t+1,i] <- L_d_juv_moy[t+1,i] - L_mu_d_juv[t+1,i]
#.....
# 2.2.2.3 stocked egg component
#.....
log(d_egg_moy[t+1,i]) <- L_d_egg_moy[t+1,i]
L_d_egg_moy[t+1,i] ~ dnorm(L_mu_d_egg[t+1,i], tau_egg_moy[ L_egg_moy[t+1,i] + 1]) I(-6.91, 1.09) #
res_egg_moy[t+1,i] <- L_d_egg_moy[t+1,i] - L_mu_d_egg[t+1,i]

# I_egg_unit = indicator of presence of incubators or not: only zone 1 and 3 concerned
# I_egg_VL = indicator for incubators in zone 1
# I_egg_LP = indicator for incubators in zone 2
# I_list_inc = indicator for each incubators loaded or not
L_mu_d_egg[t+1,i] <- equals(i,1) *
  log(
    (1- I_egg_moy[t+1,1]) +

    (s_egg * ((stock_egg[t+1,1] + stock_egg[t+1,2] + stock_egg[t+1,3] + stock_egg[t+1,4]) /
S_juv_JP[t+1,1] ))
  )
  +
  equals(i,3) *
  log(
    (1- I_egg_moy[t+1,3]) +
    (s_egg * ((stock_egg[t+1,5] + stock_egg[t+1,6]) / S_juv_JP[t+1,3] ))
  )

# getting out of the zone loop, one loop for each zones and the local densitie
# to avoid using 3 dimensions matrix
}

#-----
# 2.2.3 Successive removal fisheries
#-----
# I_site_juv_V/L/P = indicator for presence/absence of stocking on the site
# loop for sites with successive removal EF

#.....
# 2.2.3.1 zone 1 : Vichy Langeac
#.....
for (k in 1:J[t+1,1]){
  d_V[t+1,k] <- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k]
  log(d_wild_V[t+1,k]) <- L_d_wild_V[t+1,k]
  L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site) I(-6.91, 1.09)

  log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
  L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[I_site_juv_V[t+1,k] + 1]) I(-6.91, 1.09)

  lambda_N_V[t+1,k] <- d_V[t+1,k] * S_depl_V[t+1,k]

```

```

#Abundance follows a Poisson distribution
N_tot_V[t+1,k]~dpois(lambda_N_V[t+1,k])

L_p_V[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
logit(p_V[t+1,k]) <-L_p_V[t+1,k]

C_1_V[t+1,k]~dbin(p_V[t+1,k],N_tot_V[t+1,k])
N_1_V[t+1,k]<-N_tot_V[t+1,k]-C_1_V[t+1,k]
#not all sites have 2 pass, this vector show which sites does
for (h in 1:pass_2_V[t+1,k]){
  C_2_V[t+1,k]~dbin(p_V[t+1,k],N_1_V[t+1,k])
  N_2_V[t+1,k]<-N_1_V[t+1,k]-C_2_V[t+1,k]
}
}

#.....
# 2.2.3.3 zone 3 : Langeac Poutes
#.....
for (k in 1:J[t+1,3]){
  d_L[t+1,k]<- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k]
  log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
  L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,3)

  log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
  L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

  lambda_N_L[t+1,k]<-d_L[t+1,k]*S_depl_L[t+1,k]
  #Abundance follows a Poisson distribution
  N_tot_L[t+1,k]~dpois(lambda_N_L[t+1,k])

  L_p_L[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
  logit(p_L[t+1,k]) <-L_p_L[t+1,k]

  C_1_L[t+1,k]~dbin(p_L[t+1,k],N_tot_L[t+1,k])
  N_1_L[t+1,k]<-N_tot_L[t+1,k]-C_1_L[t+1,k]
  #not all sites have 2 pass, this vector show which sites does
  for (h in 1:pass_2_L[t+1,k]){
    C_2_L[t+1,k]~dbin(p_L[t+1,k],N_1_L[t+1,k])
    N_2_L[t+1,k]<-N_1_L[t+1,k]-C_2_L[t+1,k]
    #not all sites have 2 pass, this vector show which sites does
    for (m in 1:pass_3_L[t+1,k]){
      C_3_L[t+1,k]~dbin(p_L[t+1,k],N_2_L[t+1,k])
    }
  }
}
}

#.....
# 2.2.3.4 zone 4 : upstream Poutes
#.....
for (k in 1:J[t+1,4]){
  d_P[t+1,k]<- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
  log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
  L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

  log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
  L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)

```

```

lambda_N_P[t+1,k]<-d_P[t+1,k]*S_depl_P[t+1,k]
#Abundance follows a Poisson distribution
N_tot_P[t+1,k]~dpois(lambda_N_P[t+1,k])

L_p_P[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
logit(p_P[t+1,k]) <-L_p_P[t+1,k]

C_1_P[t+1,k]~dbin(p_P[t+1,k],N_tot_P[t+1,k])
N_1_P[t+1,k]<-N_tot_P[t+1,k]-C_1_P[t+1,k]
#not all sites have 2 pass, this vector show which sites does
for (h in 1:pass_2_P[t+1,k]){
  C_2_P[t+1,k]~dbin(p_P[t+1,k],N_1_P[t+1,k])
  N_2_P[t+1,k]<-N_1_P[t+1,k]-C_2_P[t+1,k]
}
}
#-----
# 2.2.4 5 min IA fisheries
#-----
#.....
# 2.2.4.1 zone 1 : Vichy Langeac
#.....
for (k in 1:K[t+1,1]){
  d_V[t+1,k]<- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k] + I_site_egg_V[t+1,k] *
d_egg_V[t+1,k]
  log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
  L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,3)

  log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
  L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[ I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)

  log(d_egg_V[t+1,k]) <- L_d_egg_V[t+1,k]
  L_d_egg_moy_V_inc[t+1,k]<- I_site_egg_V[t+1,k] * ( L_d_egg_moy[t+1,1] + log( S_juv_JP[t+1,1]) -
log(S_inc_JP[t+1,1]) )
  L_d_egg_V[t+1,k] ~ dnorm( L_d_egg_moy_V_inc[t+1,k] , tau_egg_site[I_site_egg_V[t+1,k] + 1])I(-
6.91,1.09)

  #5minute EF part
  lambda_IA_V[t+1,k]<-kappa_cut*d_V[t+1,k]
  EF_IA_V[t+1,k]~dpois(lambda_IA_V[t+1,k])
}

#.....
# 2.2.4.2 zone 2 : Alagnon
#.....
for (k in 1:K[t+1,2]){
  d_A[t+1,k]<- d_wild_A[t+1,k] + I_site_juv_A[t+1,k] * d_juv_A[t+1,k]
  log(d_wild_A[t+1,k])<-L_d_wild_A[t+1,k]
  L_d_wild_A[t+1,k] ~ dnorm( L_d_wild_moy[t+1,2] , tau_wild_site)I(-6.91,1.09)

  log(d_juv_A[t+1,k]) <- L_d_juv_A[t+1,k]
  L_d_juv_A[t+1,k] ~ dnorm( L_d_juv_moy[t+1,2] , tau_juv_site[ I_site_juv_A[t+1,k] + 1])I(-6.91,1.09)

  #5minute EF part
  lambda_IA_A[t+1,k]<-kappa_cut*d_A[t+1,k]
  EF_IA_A[t+1,k]~dpois(lambda_IA_A[t+1,k])
}

```

```

#.....
# 2.2.4.3 zone 3 : Langeac Poutes
#.....
for (k in 1:K[t+1,3]){
  d_L[t+1,k]<- d_wild_L[t+1,k] + l_site_juv_L[t+1,k] * d_juv_L[t+1,k] + l_site_egg_L[t+1,k] *
d_egg_L[t+1,k]
  log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
  L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)l(-6.91,1.09)

  log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
  L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ l_site_juv_L[t+1,k] + 1])l(-6.91,1.09)

  log(d_egg_L[t+1,k]) <- L_d_egg_L[t+1,k]
  L_d_egg_moy_L_inc[t+1,k]<- l_site_egg_L[t+1,k] * ( L_d_egg_moy[t+1,3] + log( S_juv_JP[t+1,3]) -
log(S_inc_JP[t+1,3]) )
  L_d_egg_L[t+1,k] ~ dnorm( L_d_egg_moy_L_inc[t+1,k] , tau_egg_site[l_site_egg_L[t+1,k] + 1])l(-
6.91,1.09)

  #5minute EF part
  lambda_IA_L[t+1,k]<-kappa_cut*d_L[t+1,k]
  EF_IA_L[t+1,k]~dpois(lambda_IA_L[t+1,k])
}
#.....
# 2.2.4.3 zone 4 : upstream Poutes
#.....
for (k in 1:K[t+1,4]){
  d_P[t+1,k]<- d_wild_P[t+1,k] + l_site_juv_P[t+1,k] * d_juv_P[t+1,k]
  log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
  L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)l(-6.91,1.09)

  log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
  L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ l_site_juv_P[t+1,k] + 1])l(-6.91,1.09)

  #5minute EF part
  lambda_IA_P[t+1,k]<-kappa_cut*d_P[t+1,k]
  EF_IA_P[t+1,k]~dpois(lambda_IA_P[t+1,k])
}
}

```

```

#####
# 3. Change in redd count methodology #
#####
for (t in 23:30){
  #=====
  # 3.1 Redd/Spawners part
  #=====
  logit(p_alagnon[t])<-L_p_alagnon[t]
  logit(p_langeac[t])<- L_p_langeac[t]
  logit(p_poutes[t])<- L_p_poutes[t]
  pool_juv[t]<-s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 * smolts_tot[t] )
  L_mu_Vichy_nm[t]<-log(s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 *
smolts_tot[t] )) + level_s * l_surv[t]
  mean_y_surv[t] <- s_juv2ad * exp(level_s * l_surv[t])
}

```



```

# max is only added for the year we only have a minimum figure at vichy
temp[t]<-max(tot_C[t] + S_stocking[t]+2,min_N_V[t])

min_N_1[t]<-max(N[t,4]+2,temp[t] +2)+S_stocking[t]
N[t,1]~dlnorm(L_mu_Vichy_nm[t],tau_vichy)l(min_N_1[t],15000)
#without fish caught for breeding or rod catches
N_corrected[t] <- N[t,1] - tot_C[t] - S_stocking[t]
N_corr[t]<-N_corrected[t] - N[t,2]
res_Vichy[t] <- log(N[t,1]) - L_mu_Vichy_nm[t]

max_N_langeac[t]<- N_corr[t] - 1
min_L_P[t]<-max(min_L[t], N[t,4]+1)
N[t,2]~dbin(p_alagnon[t],N_corrected[t]) #Pas de l car on pourrait avoir 0 sur alagnon et tout sur allier
une année donnée
N[t,3]~dbin(p_langeac[t],N_corr[t])l(min_L_P[t],)

max_N_poutes[t]<-N[t,3]-1
N[t,4]~dbin(p_poutes[t],N[t,3])
#-----
# 3.1.1 Number of potential spawners
#-----
#mu_S_ts[t,1]<- N[t,1] - N[t,2] - S_stocking[t]
#mu_S_ts[t,2]<- N[t,2]-N[t,3]
#test[t]<-mu_S_ts[t,1]-S_ts[t,1]

S_ts[t,1]<- max(N[t,1] - N[t,2] - N[t,3] - S_stocking[t] - tot_C[t] ,1)
S_ts[t,2]<- max( N[t,2],1)
S_ts[t,3]<- max( N[t,3]-N[t,4],1)
S_ts[t,4]<-max( N[t,4],1)

#ratio_S[t,1] <- S_ts[t,1] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,2] <- S_ts[t,2] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,3] <- S_ts[t,3] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,4] <- S_ts[t,4] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])

#=====
# 3.2 Loop for zones (1= Vichy-Langeac, 2= Alagnon, 3= Langeac-Poutès, 4= upstream Poutès)
#=====
for (i in 1:4){
#-----
# 3.2.1 Redd/Spawners part
#-----
#.....
# 3.2.1.1 Estimation of the spawners
#.....
R[t,i]~dpois(lambda[t,i])
lambda[t,i] <- S_ts[t,i] *zone_effect[t,i]* hel_effect[2] *p_area[t,i]
res_R[t,i]<-(R[t,i]-lambda[t,i])/sqrt(lambda[t,i])
#.....
# 3.2.1.2 Cut of all parameters
#.....
lambda_cut[t,i]<-cut(lambda[t,i])
R_rep[t,i]~dpois(lambda_cut[t,i])

#S_counter[t,i]<-R[t,i] / (zone_effect[t,i] *p_area[t,i])

```

```

#chisq_disc_R[t,i]<- (R[t,i]-lambda[t,i]) * (R[t,i]-lambda[t,i]) / (lambda[t,i])
#chisq_disc_R_rep[t,i]<- (R_rep[t,i]-lambda[t,i]) * (R_rep[t,i]-lambda[t,i]) / (lambda[t,i])
#-----
# 3.2.2 Juvenile production
#-----
# l_juv_moy = indicator for stocking of 0+ or not
# l_egg_moy = indicator for stocking of eggs or not
#d_tot_moy without taking into account area for the stocked juveniles (data only from year 31)
d_tot_moy[t+1,i] <- d_wild_moy[t+1,i] + l_juv_moy[t+1,i] * d_juv_moy[t+1,i] + l_egg_moy[t+1,i] *
d_egg_moy_surf[t+1,i]
Juv[t+1,i] <- d_tot_moy[t+1,i]*S_juv_JP[t+1,i]
d_egg_moy_surf[t+1,i] <- d_egg_moy[t+1,i]
#.....
# 3.2.2.1 Wild component
#.....
log(d_wild_moy[t+1,i]) <- L_d_wild_moy[t+1,i]
L_d_wild_moy[t+1,i] ~ dnorm(L_mu_d_wild[t+1,i],tau_wild_moy)I(-6.91,1.09) #<-
L_mu_d_wild[t+1,i] #
L_mu_d_wild[t+1,i] <- log((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd * (S_ts[t,i]/S_juv_JP[t,i]))) +
nu_wild[i]
res_wild_moy[t+1,i] <- L_d_wild_moy[t+1,i] - L_mu_d_wild[t+1,i]
#.....
# 3.2.2.2 Stocked juvenile component
#.....
log(d_juv_moy[t+1,i]) <- L_d_juv_moy[t+1,i]
L_d_juv_moy[t+1,i] ~ dnorm(L_mu_d_juv[t+1,i],tau_juv_moy[l_juv_moy[t+1,i]+1])I(1,0.9)

# We recalculate the Rmax "available" to stocked 0+ by subtracting wild 0+ density and stocked
eggs density
# to the total Rmax of the density dependence relationship
Rmax_juv_temp[t+1,i] <- ( Rmax - ((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd *
(S_ts[t,i]/S_juv_JP[t,i]))) ) * exp(nu_wild[i])
Rmax_juv[t+1,i] <- max(Rmax_juv_temp[t+1,i],0.000001)
beta_dd_juv[t+1,i] <- 1 / Rmax_juv[t+1,i]
L_mu_d_juv[t+1,i] <- l_juv_moy[t+1,i] * log( (stock_juv[t+1,i]/S_juv_JP[t+1,i]) /
(alpha_dd_juv/exp(nu_wild[i]) + beta_dd_juv[t+1,i] * (stock_juv[t+1,i]/S_juv_JP[t+1,i])))
res_juv_moy[t+1,i] <- L_d_juv_moy[t+1,i] - L_mu_d_juv[t+1,i]
#.....
# 3.2.2.3 Stocked egg component
#.....
log(d_egg_moy[t+1,i]) <- L_d_egg_moy[t+1,i]
L_d_egg_moy[t+1,i] ~ dnorm(L_mu_d_egg[t+1,i],tau_egg_moy[l_egg_moy[t+1,i] +1])I(-6.91,1.09) #
res_egg_moy[t+1,i] <- L_d_egg_moy[t+1,i] - L_mu_d_egg[t+1,i]

# l_egg_unit = indicator of presence of incubators or not: only zone 1 and 2 concerned
# l_egg_VL = indicator for incubators in zone 1
# l_egg_LP = indicator for incubators in zone 2
# l_list_inc = indicator for each incubators loaded or not
L_mu_d_egg[t+1,i] <- equals(i,1) *
log(
(1- l_egg_moy[t+1,1]) +

(s_egg * ((stock_egg[t+1,1] + stock_egg[t+1,2] + stock_egg[t+1,3] + stock_egg[t+1,4]) /
S_juv_JP[t+1,1] ))
)
+

```

```

        equals(i,3) *
        log(
        (1- l_egg_moy[t+1,3]) +
        (s_egg * ((stock_egg[t+1,5] +stock_egg[t+1,6]) / S_juv_JP[t+1,2] ))
        )
# getting out of the zone loop, one loop for each zones and the local densitie
# to avoid using 3 dimensions matrix
}

#-----
# 3.2.3 Successive removal fisheries
#-----
# l_site_juv_V/L/P = indicator for presence/absence of stocking on the site
# loop for sites with successive removal EF

#.....
# 3.2.3.1 zone 1 : Vichy Langeac
#.....
for (k in 1:J[t+1,1]){
  d_V[t+1,k]<- d_wild_V[t+1,k] + l_site_juv_V[t+1,k] * d_juv_V[t+1,k]
  log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
  L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,1.09)

  log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
  L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[l_site_juv_V[t+1,k] + 1])I(-6.91,1.09)

  lambda_N_V[t+1,k]<-d_V[t+1,k]*S_depl_V[t+1,k]
  #Abundance follows a Poisson distribution
  N_tot_V[t+1,k]~dpois(lambda_N_V[t+1,k])

  L_p_V[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
  logit(p_V[t+1,k]) <-L_p_V[t+1,k]

  C_1_V[t+1,k]~dbin(p_V[t+1,k],N_tot_V[t+1,k])
  N_1_V[t+1,k]<-N_tot_V[t+1,k]-C_1_V[t+1,k]
  #not all sites have 2 pass, this vector show which sites does
  for (h in 1:pass_2_V[t+1,k]){
    C_2_V[t+1,k]~dbin(p_V[t+1,k],N_1_V[t+1,k])
    N_2_V[t+1,k]<-N_1_V[t+1,k]-C_2_V[t+1,k]
  }
}

#.....
# 3.2.3.3 zone 3 : Langeac Poutes
#.....
for (k in 1:J[t+1,3]){
  d_L[t+1,k]<- d_wild_L[t+1,k] + l_site_juv_L[t+1,k] * d_juv_L[t+1,k]
  log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
  L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,3)

  log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
  L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ l_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

  lambda_N_L[t+1,k]<-d_L[t+1,k]*S_depl_L[t+1,k]
  #Abundance follows a Poisson distribution
  N_tot_L[t+1,k]~dpois(lambda_N_L[t+1,k])

```

```

L_p_L[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
logit(p_L[t+1,k]) <-L_p_L[t+1,k]

C_1_L[t+1,k]~dbin(p_L[t+1,k],N_tot_L[t+1,k])
N_1_L[t+1,k]<-N_tot_L[t+1,k]-C_1_L[t+1,k]
#not all sites have 2 pass, this vector show which sites does
for (h in 1:pass_2_L[t+1,k]){
  C_2_L[t+1,k]~dbin(p_L[t+1,k],N_1_L[t+1,k])
  N_2_L[t+1,k]<-N_1_L[t+1,k]-C_2_L[t+1,k]
  #not all sites have 2 pass, this vector show which sites does
  for (m in 1:pass_3_L[t+1,k]){
    C_3_L[t+1,k]~dbin(p_L[t+1,k],N_2_L[t+1,k])
  }
}
}
#.....
# 3.2.3.4 zone 4 : upstream Poutes
#.....
for (k in 1:J[t+1,4]){
  d_P[t+1,k]<- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
  log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
  L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

  log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
  L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)

  lambda_N_P[t+1,k]<-d_P[t+1,k]*S_depl_P[t+1,k]
  #Abundance follows a Poisson distribution
  N_tot_P[t+1,k]~dpois(lambda_N_P[t+1,k])

  L_p_P[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
  logit(p_P[t+1,k]) <-L_p_P[t+1,k]

  C_1_P[t+1,k]~dbin(p_P[t+1,k],N_tot_P[t+1,k])
  N_1_P[t+1,k]<-N_tot_P[t+1,k]-C_1_P[t+1,k]
  #not all sites have 2 pass, this vector show which sites does
  for (h in 1:pass_2_P[t+1,k]){
    C_2_P[t+1,k]~dbin(p_P[t+1,k],N_1_P[t+1,k])
    N_2_P[t+1,k]<-N_1_P[t+1,k]-C_2_P[t+1,k]
  }
}
#-----
# 3.2.4 5 min IA fisheries
#-----
#.....
# 3.2.4.1 zone 1 : Vichy Langeac
#.....
for (k in 1:K[t+1,1]){
  d_V[t+1,k]<- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k] + I_site_egg_V[t+1,k] *
d_egg_V[t+1,k]
  log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
  L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,3)

  log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
  L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[ I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)

```

```

log(d_egg_V[t+1,k]) <- L_d_egg_V[t+1,k]
L_d_egg_moy_V_inc[t+1,k] <- I_site_egg_V[t+1,k] * ( L_d_egg_moy[t+1,1] + log( S_juv_JP[t+1,1]) -
log(S_inc_JP[t+1,1]) )
L_d_egg_V[t+1,k] ~ dnorm( L_d_egg_moy_V_inc[t+1,k] , tau_egg_site[I_site_egg_V[t+1,k] + 1])|(-
6.91,1.09)

#5minute EF part
lambda_IA_V[t+1,k] <- kappa_cut*d_V[t+1,k]          #kappa_cut*pow(d_V[t+1,k],eta_cut)
EF_IA_V[t+1,k] ~ dpois(lambda_IA_V[t+1,k])
}

#.....
# 3.2.4.2 zone 2 : Alagnon
#.....
for (k in 1:K[t+1,2]){
d_A[t+1,k] <- d_wild_A[t+1,k] + I_site_juv_A[t+1,k] * d_juv_A[t+1,k]
log(d_wild_A[t+1,k]) <- L_d_wild_A[t+1,k]
L_d_wild_A[t+1,k] ~ dnorm( L_d_wild_moy[t+1,2] , tau_wild_site)|(-6.91,1.09)

log(d_juv_A[t+1,k]) <- L_d_juv_A[t+1,k]
L_d_juv_A[t+1,k] ~ dnorm( L_d_juv_moy[t+1,2] , tau_juv_site[ I_site_juv_A[t+1,k] + 1])|(-6.91,1.09)

#5minute EF part
lambda_IA_A[t+1,k] <- kappa_cut*d_A[t+1,k]
EF_IA_A[t+1,k] ~ dpois(lambda_IA_A[t+1,k])
}

#.....
# 3.2.4.3 zone 3 : Langeac Poutes
#.....
for (k in 1:K[t+1,3]){
d_L[t+1,k] <- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k] + I_site_egg_L[t+1,k] *
d_egg_L[t+1,k]
log(d_wild_L[t+1,k]) <- L_d_wild_L[t+1,k]
L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)|(-6.91,1.09)

log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])|(-6.91,1.09)

log(d_egg_L[t+1,k]) <- L_d_egg_L[t+1,k]
L_d_egg_moy_L_inc[t+1,k] <- I_site_egg_L[t+1,k] * ( L_d_egg_moy[t+1,3] + log( S_juv_JP[t+1,3]) -
log(S_inc_JP[t+1,3]) )
L_d_egg_L[t+1,k] ~ dnorm( L_d_egg_moy_L_inc[t+1,k] , tau_egg_site[I_site_egg_L[t+1,k] + 1])|(-
6.91,1.09)

#5minute EF part
lambda_IA_L[t+1,k] <- kappa_cut*d_L[t+1,k]
EF_IA_L[t+1,k] ~ dpois(lambda_IA_L[t+1,k])
}
#.....
# 3.2.4.4 zone 4 : upstream Poutes
#.....
for (k in 1:K[t+1,4]){
d_P[t+1,k] <- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
log(d_wild_P[t+1,k]) <- L_d_wild_P[t+1,k]

```

```

L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)

#5minute EF part
lambda_IA_P[t+1,k]<-kappa_cut*d_P[t+1,k]
EF_IA_P[t+1,k]~dpois(lambda_IA_P[t+1,k])
}
}

#####
# 4. Take into account the area used for stocked juveniles #
#####
for (t in 31:T-1){
  #####
  # 4.1 Redd/Spawners part
  #####
  logit(p_alagnon[t])<-L_p_alagnon[t]
  logit(p_langeac[t])<- L_p_langeac[t]
  logit(p_poutes[t])<- L_p_poutes[t]
  pool_juv[t]<-s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 * smolts_tot[t] )
  L_mu_Vichy_nm[t]<-log(s_juv2ad * Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 *
smolts_tot[t] )) + level_s * I_surv[t]
  mean_y_surv[t] <- s_juv2ad * exp(level_s * I_surv[t])

  # max is only added for the year we only have a minimum figure at vichy
  temp[t]<-max(tot_C[t] + S_stocking[t]+2,min_N_V[t])

  min_N_1[t]<-max(N[t,4]+2,temp[t] +2)+S_stocking[t]
  N[t,1]~dlnorm(L_mu_Vichy_nm[t],tau_vichy)I(min_N_1[t],15000)
  #without fish caught for breeding or rod catches
  N_corrected[t] <- N[t,1] - tot_C[t] - S_stocking[t]
  N_corr[t]<-N_corrected[t] - N[t,2]
  res_Vichy[t] <- log(N[t,1]) - L_mu_Vichy_nm[t]

  max_N_langeac[t]<- N_corr[t] - 1
  min_L_P[t]<-max(min_L[t], N[t,4]+1)
  N[t,2]~dbin(p_alagnon[t],N_corrected[t]) #Pas de 1 car on pourrait avoir 0 sur alagnon et tout sur allier
  une année donnée
  N[t,3]~dbin(p_langeac[t],N_corr[t])I(min_L_P[t],)

  max_N_poutes[t]<-N[t,3]-1
  N[t,4]~dbin(p_poutes[t],N[t,3])
  #-----
  # 4.1.1 Number of potential spawners
  #-----
  #mu_S_ts[t,1]<- N[t,1] - N[t,2] - S_stocking[t]
  #mu_S_ts[t,2]<- N[t,2]-N[t,3]
  #test[t]<-mu_S_ts[t,1]-S_ts[t,1]

  S_ts[t,1]<- max(N[t,1] - N[t,2] - N[t,3] - S_stocking[t] - tot_C[t] ,1)
  S_ts[t,2]<- max( N[t,2],1)
  S_ts[t,3]<- max( N[t,3]-N[t,4],1)
  S_ts[t,4]<-max( N[t,4],1)

```

```

#ratio_S[t,1] <- S_ts[t,1] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,2] <- S_ts[t,2] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,3] <- S_ts[t,3] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,4] <- S_ts[t,4] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])

#=====
# 4.2 Loop for zones (1= Vichy-Langeac, 2= Alagnon, 3= Langeac-Poutès, 4= upstream Poutès)
#=====

for (i in 1:4){
#-----
# 4.2.1 Redd/Spawners part
#-----
#.....
# 4.2.1.1 Estimation of the spawners
#.....
R[t,i]~dpois(lambda[t,i])
lambda[t,i] <- S_ts[t,i] *zone_effect[t,i]* hel_effect[2] *p_area[t,i]
res_R[t,i]<-(R[t,i]-lambda[t,i])/sqrt(lambda[t,i])
#.....
# 4.2.1.2 Cut of all parameters
#.....
lambda_cut[t,i]<-cut(lambda[t,i])
R_rep[t,i]~dpois(lambda_cut[t,i])

#S_counter[t,i]<-R[t,i] / (zone_effect[t,i] *p_area[t,i])
#chisq_disc_R[t,i]<- (R[t,i]-lambda[t,i]) * (R[t,i]-lambda[t,i]) / (lambda[t,i])
#chisq_disc_R_rep[t,i]<- (R_rep[t,i]-lambda[t,i]) * (R_rep[t,i]-lambda[t,i]) / (lambda[t,i])
#-----
# 4.2.2 Juvenile production
#-----
# l_juv_moy = indicator for stocking of 0+ or not
# l_egg_moy = indicator for stocking of eggs or not
#d_tot_moy with taking into account area for the stocked juveniles (data only from year 31)
d_tot_moy[t+1,i] <- d_wild_moy[t+1,i] + l_juv_moy[t+1,i] *
d_juv_moy[t+1,i]*S_juv_JP_dev[t+1,i]/S_juv_JP[t+1,i] + l_egg_moy[t+1,i] * d_egg_moy_surf[t+1,i]
Juv[t+1,i] <- d_tot_moy[t+1,i]*S_juv_JP[t+1,i]
d_egg_moy_surf[t+1,i] <- d_egg_moy[t+1,i]
#.....
# 4.2.2.1 Wild component
#.....
log(d_wild_moy[t+1,i]) <- L_d_wild_moy[t+1,i]
L_d_wild_moy[t+1,i] ~ dnorm(L_mu_d_wild[t+1,i],tau_wild_moy)I(-6.91,1.09) #<-
L_mu_d_wild[t+1,i] #
L_mu_d_wild[t+1,i] <- log((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd * (S_ts[t,i]/S_juv_JP[t,i]))) +
nu_wild[i]
res_wild_moy[t+1,i] <- L_d_wild_moy[t+1,i] - L_mu_d_wild[t+1,i]
#.....
# 4.2.2.2 Stocked juvenile component
#.....
log(d_juv_moy[t+1,i]) <- L_d_juv_moy[t+1,i]
L_d_juv_moy[t+1,i] ~ dnorm(L_mu_d_juv[t+1,i],tau_juv_moy[l_juv_moy[t+1,i]+1])I(,1.09)

# We recalculate the Rmax "available" to stocked 0+ by subtracting wild 0+ density and stocked
eggs density
# to the total Rmax of the density dependence relationship

```

```

Rmax_juv_temp[t+1,i] <- ( Rmax - ((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd *
(S_ts[t,i]/S_juv_JP[t,i]))) ) * exp(nu_wild[i])
Rmax_juv[t+1,i] <- max(Rmax_juv_temp[t+1,i],0.000001)
beta_dd_juv[t+1,i] <- 1 / Rmax_juv[t+1,i]
L_mu_d_juv[t+1,i] <- l_juv_moy[t+1,i] * log( (stock_juv[t+1,i]/S_juv_JP[t+1,i]) /
(alpha_dd_juv/exp(nu_wild[i]) + beta_dd_juv[t+1,i] * (stock_juv[t+1,i]/S_juv_JP[t+1,i])))
res_juv_moy[t+1,i] <- L_d_juv_moy[t+1,i] - L_mu_d_juv[t+1,i]
#.....
# 4.2.2.3 Stocked egg component
#.....
log(d_egg_moy[t+1,i]) <- L_d_egg_moy[t+1,i]
L_d_egg_moy[t+1,i] ~ dnorm(L_mu_d_egg[t+1,i],tau_egg_moy[ l_egg_moy[t+1,i] +1])l(-6.91,1.09) #
res_egg_moy[t+1,i] <- L_d_egg_moy[t+1,i] - L_mu_d_egg[t+1,i]

# l_egg_unit = indicator of presence of incubators or not: only zone 1 and 2 concerned
# l_egg_VL = indicator for incubators in zone 1
# l_egg_LP = indicator for incubators in zone 2
# l_list_inc = indicator for each incubators loaded or not
L_mu_d_egg[t+1,i] <- equals(i,1) *
log(
(1- l_egg_moy[t+1,1]) +

(s_egg * ((stock_egg[t+1,1] + stock_egg[t+1,2] + stock_egg[t+1,3] + stock_egg[t+1,4]) /
S_juv_JP[t+1,1] ))
)
+
equals(i,3) *
log(
(1- l_egg_moy[t+1,3]) +
(s_egg * ((stock_egg[t+1,5] +stock_egg[t+1,6]) / S_juv_JP[t+1,2] ))
)
# getting out of the zone loop, one loop for each zones and the local densitie
# to avoid using 3 dimensions matrix
}

#-----
# 4.2.3 Successive removal fisheries
#-----
# l_site_juv_V/L/P = indicator for presence/absence of stocking on the site
# loop for sites with successive removal EF

#.....
# 4.2.3.1 zone 1 : Vichy Langeac
#.....
for (k in 1:J[t+1,1]){
d_V[t+1,k]<- d_wild_V[t+1,k] + l_site_juv_V[t+1,k] * d_juv_V[t+1,k]
log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)l(-6.91,1.09)

log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[l_site_juv_V[t+1,k] + 1])l(-6.91,1.09)

lambda_N_V[t+1,k]<-d_V[t+1,k]*S_depl_V[t+1,k]
#Abundance follows a Poisson distribution
N_tot_V[t+1,k]~dpois(lambda_N_V[t+1,k])

L_p_V[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)

```



```

logit(p_V[t+1,k]) <- L_p_V[t+1,k]

C_1_V[t+1,k]~dbin(p_V[t+1,k],N_tot_V[t+1,k])
N_1_V[t+1,k]<-N_tot_V[t+1,k]-C_1_V[t+1,k]
#not all sites have 2 pass, this vector show which sites does
for (h in 1:pass_2_V[t+1,k]){
  C_2_V[t+1,k]~dbin(p_V[t+1,k],N_1_V[t+1,k])
  N_2_V[t+1,k]<-N_1_V[t+1,k]-C_2_V[t+1,k]
}
}

#.....
# 4.2.3.3 zone 3 : Langeac Poutes
#.....
for (k in 1:J[t+1,3]){
  d_L[t+1,k]<- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k]
  log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
  L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,3)

  log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
  L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

  lambda_N_L[t+1,k]<-d_L[t+1,k]*S_depl_L[t+1,k]
  #Abundance follows a Poisson distribution
  N_tot_L[t+1,k]~dpois(lambda_N_L[t+1,k])

  L_p_L[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
  logit(p_L[t+1,k]) <-L_p_L[t+1,k]

  C_1_L[t+1,k]~dbin(p_L[t+1,k],N_tot_L[t+1,k])
  N_1_L[t+1,k]<-N_tot_L[t+1,k]-C_1_L[t+1,k]
  #not all sites have 2 pass, this vector show which sites does
  for (h in 1:pass_2_L[t+1,k]){
    C_2_L[t+1,k]~dbin(p_L[t+1,k],N_1_L[t+1,k])
    N_2_L[t+1,k]<-N_1_L[t+1,k]-C_2_L[t+1,k]
    #not all sites have 2 pass, this vector show which sites does
    for (m in 1:pass_3_L[t+1,k]){
      C_3_L[t+1,k]~dbin(p_L[t+1,k],N_2_L[t+1,k])
    }
  }
}
}

#.....
# 4.2.3.4 zone 4 : upstream Poutes
#.....
for (k in 1:J[t+1,4]){
  d_P[t+1,k]<- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
  log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
  L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

  log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
  L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)

  lambda_N_P[t+1,k]<-d_P[t+1,k]*S_depl_P[t+1,k]
  #Abundance follows a Poisson distribution
  N_tot_P[t+1,k]~dpois(lambda_N_P[t+1,k])

```

```

L_p_P[t+1,k]~dnorm(L_mu_p_cut,L_tau_p_cut)
logit(p_P[t+1,k]) <-L_p_P[t+1,k]

C_1_P[t+1,k]~dbin(p_P[t+1,k],N_tot_P[t+1,k])
N_1_P[t+1,k]<-N_tot_P[t+1,k]-C_1_P[t+1,k]
#not all sites have 2 pass, this vector show which sites does
for (h in 1:pass_2_P[t+1,k]){
  C_2_P[t+1,k]~dbin(p_P[t+1,k],N_1_P[t+1,k])
  N_2_P[t+1,k]<-N_1_P[t+1,k]-C_2_P[t+1,k]
}
}
#-----
# 4.2.4 5 min IA fisheries
#-----
#.....
# 4.2.4.1 zone 1 : Vichy Langeac
#.....
for (k in 1:K[t+1,1]){
  d_V[t+1,k]<- d_wild_V[t+1,k] + I_site_juv_V[t+1,k] * d_juv_V[t+1,k] + I_site_egg_V[t+1,k] *
d_egg_V[t+1,k]
  log(d_wild_V[t+1,k])<-L_d_wild_V[t+1,k]
  L_d_wild_V[t+1,k] ~ dnorm( L_d_wild_moy[t+1,1] , tau_wild_site)I(-6.91,3)

  log(d_juv_V[t+1,k]) <- L_d_juv_V[t+1,k]
  L_d_juv_V[t+1,k] ~ dnorm( L_d_juv_moy[t+1,1] , tau_juv_site[ I_site_juv_V[t+1,k] + 1])I(-6.91,1.09)

  log(d_egg_V[t+1,k]) <- L_d_egg_V[t+1,k]
  L_d_egg_moy_V_inc[t+1,k]<- I_site_egg_V[t+1,k] * ( L_d_egg_moy[t+1,1] + log( S_juv_JP[t+1,1]) -
log(S_inc_JP[t+1,1]) )
  L_d_egg_V[t+1,k] ~ dnorm( L_d_egg_moy_V_inc[t+1,k] , tau_egg_site[I_site_egg_V[t+1,k] + 1])I(-
6.91,1.09)

  #5minute EF part
  lambda_IA_V[t+1,k]<-kappa_cut*d_V[t+1,k]          #kappa_cut*pow(d_V[t+1,k],eta_cut)
  EF_IA_V[t+1,k]~dpois(lambda_IA_V[t+1,k])
}

#.....
# 4.2.4.2 zone 2 : Alagnon
#.....
for (k in 1:K[t+1,2]){
  d_A[t+1,k]<- d_wild_A[t+1,k] + I_site_juv_A[t+1,k] * d_juv_A[t+1,k]
  log(d_wild_A[t+1,k])<-L_d_wild_A[t+1,k]
  L_d_wild_A[t+1,k] ~ dnorm( L_d_wild_moy[t+1,2] , tau_wild_site)I(-6.91,1.09)

  log(d_juv_A[t+1,k]) <- L_d_juv_A[t+1,k]
  L_d_juv_A[t+1,k] ~ dnorm( L_d_juv_moy[t+1,2] , tau_juv_site[ I_site_juv_A[t+1,k] + 1])I(-6.91,1.09)

  #5minute EF part
  lambda_IA_A[t+1,k]<-kappa_cut*d_A[t+1,k]
  EF_IA_A[t+1,k]~dpois(lambda_IA_A[t+1,k])
}

#.....
# 4.2.4.3 zone 3 : Langeac Poutes

```

```

#.....
for (k in 1:K[t+1,3]){
  d_L[t+1,k]<- d_wild_L[t+1,k] + I_site_juv_L[t+1,k] * d_juv_L[t+1,k] + I_site_egg_L[t+1,k] *
d_egg_L[t+1,k]
  log(d_wild_L[t+1,k])<-L_d_wild_L[t+1,k]
  L_d_wild_L[t+1,k] ~ dnorm( L_d_wild_moy[t+1,3] , tau_wild_site)I(-6.91,1.09)

  log(d_juv_L[t+1,k]) <- L_d_juv_L[t+1,k]
  L_d_juv_L[t+1,k] ~ dnorm( L_d_juv_moy[t+1,3] , tau_juv_site[ I_site_juv_L[t+1,k] + 1])I(-6.91,1.09)

  log(d_egg_L[t+1,k]) <- L_d_egg_L[t+1,k]
  L_d_egg_moy_L_inc[t+1,k]<- I_site_egg_L[t+1,k] * ( L_d_egg_moy[t+1,3] + log( S_juv_JP[t+1,3]) -
log(S_inc_JP[t+1,3]) )
  L_d_egg_L[t+1,k] ~ dnorm( L_d_egg_moy_L_inc[t+1,k] , tau_egg_site[I_site_egg_L[t+1,k] + 1])I(-
6.91,1.09)

  #5minute EF part
  lambda_IA_L[t+1,k]<-kappa_cut*d_L[t+1,k]
  EF_IA_L[t+1,k]~dpois(lambda_IA_L[t+1,k])
}

#.....
# 4.2.4.4 zone 4 : upstream Poutes
#.....
for (k in 1:K[t+1,4]){
  d_P[t+1,k]<- d_wild_P[t+1,k] + I_site_juv_P[t+1,k] * d_juv_P[t+1,k]
  log(d_wild_P[t+1,k])<-L_d_wild_P[t+1,k]
  L_d_wild_P[t+1,k] ~ dnorm( L_d_wild_moy[t+1,4] , tau_wild_site)I(-6.91,1.09)

  log(d_juv_P[t+1,k]) <- L_d_juv_P[t+1,k]
  L_d_juv_P[t+1,k] ~ dnorm( L_d_juv_moy[t+1,4] , tau_juv_site[ I_site_juv_P[t+1,k] + 1])I(-6.91,1.09)

  #5minute EF part
  lambda_IA_P[t+1,k]<-kappa_cut*d_P[t+1,k]
  EF_IA_P[t+1,k]~dpois(lambda_IA_P[t+1,k])
}
}

#####
# 5. Just the last year to estimate spawners #
#####
for (t in T:T){
  #=====
  # 5.1 Redd/Spawners part
  #=====
  logit(p_alagnon[t])<-L_p_alagnon[t]
  logit(p_langeac[t])<- L_p_langeac[t]
  logit(p_poutes[t])<- L_p_poutes[t]
  pool_juv[t]<-s_juv2ad* Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 * smolts_tot[t] )
  L_mu_Vichy_nm[t]<-log(s_juv2ad *Juv_tot_system[t] + s_smolt * (0.5 * smolts_tot[t+1] + 0.5 *
smolts_tot[t] )) + level_s *I_surv[t]

  min_N_1[t]<-max(N[t,4]+2,tot_C[t] +2)+S_stocking[t]
  N[t,1]~dlnorm(L_mu_Vichy_nm[t],tau_vichy)I(min_N_1[t],15000)
  #without fish caught for breeding
  N_corrected[t]<-N[t,1]-S_stocking[t]

```

```

N_corr[t]<-N_corrected[t] - N[t,2]
res_Vichy[t] <- log(N[t,1]) - L_mu_Vichy_nm[t]

max_N_langeac[t]<- N_corr[t] -1
min_L_P[t]<-N[t,4]+1

N[t,2]~dbin(p_alagnon[t],N_corrected[t]) #Pas de 1 car on pourrait avoir 0 sur alagnon et tout sur
allier une année donnée
N[t,3]~dbin(p_langeac[t],N_corr[t])l(min_L_P[t],)

max_N_poutes[t]<-N[t,3]-1
N[t,4]~dbin(p_poutes[t],N[t,3])

#-----
# 5.1.1 Number of potential spawners
#-----
S_ts[t,1]<- max(N[t,1] - N[t,2] - N[t,3] - S_stocking[t] - tot_C[t] ,1)
S_ts[t,2]<- max( N[t,2],1)
S_ts[t,3]<- max( N[t,3]-N[t,4],1)
S_ts[t,4]<-max( N[t,4],1)

#ratio_S[t,1] <- S_ts[t,1] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,2] <- S_ts[t,2] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,3] <- S_ts[t,3] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])
#ratio_S[t,4] <- S_ts[t,4] / ( S_ts[t,1] + S_ts[t,2] + S_ts[t,3] + S_ts[t,4])

#=====
# 5.2 Loop for zones (1= Vichy-Langeac, 2= Langeac-Poutès, 3= upstream Poutès)
#=====
for (i in 1:4){
#-----
# 5.2.1 Redd/Spawners part
#-----
#.....
# 5.2.1.1 Estimation of the spawners
#.....
R[t,i]~dpois(lambda[t,i])
lambda[t,i] <- S_ts[t,i] *zone_effect[t,i] * hel_effect[2]*p_area[t,i]
res_R[t,i]<-(R[t,i]-lambda[t,i])/sqrt(lambda[t,i])
#.....
# 5.2.1.2 Cut of all parameters
#.....
lambda_cut[t,i]<-cut(lambda[t,i])
R_rep[t,i]~dpois(lambda_cut[t,i])

#S_counter[t,i]<-R[t,i] / (zone_effect[t,i] *p_area[t,i])
#chisq_disc_R[t,i]<- (R[t,i]-lambda[t,i]) * (R[t,i]-lambda[t,i]) / (lambda[t,i])
#chisq_disc_R_rep[t,i]<- (R_rep[t,i]-lambda[t,i]) * (R_rep[t,i]-lambda[t,i]) / (lambda[t,i])
#-----
# 5.2.2 Juvenile production (wild only)
#-----
d_tot_moy[t+1,i] <- d_wild_moy[t+1,i]
Juv[t+1,i] <- d_tot_moy[t+1,i]*S_juv_JP[t+1,i]
#.....
# 5.2.2.1 Wild component
#.....

```

```

        log(d_wild_moy[t+1,i]) <- L_d_wild_moy[t+1,i]
        L_d_wild_moy[t+1,i] ~ dnorm(L_mu_d_wild[t+1,i],tau_wild_moy)l(-6.91,1.09)
        L_mu_d_wild[t+1,i] <- log((S_ts[t,i]/S_juv_JP[t,i]) / (alpha_dd + beta_dd * (S_ts[t,i]/S_juv_JP[t,i]))) +
nu_wild[i]
        res_wild_moy[t+1,i] <- L_d_wild_moy[t+1,i] - L_mu_d_wild[t+1,i]
    }
}

#### END MODEL BRACKET
}

```